Cheker Neffati

Montréal, Québec • cheker.neffati@etsmtl.ca • linkedin.com/in/cheker-neffati

M.Sc.A. in Automated Production Engineering • Deep learning for time series
Cumulative GPA 4.12/4.30 • C-PILOT project (CRIAQ, NSERC, Bombardier) • 3 scientific communications

PROFILE

Master's student (thesis option) in Automated Production Engineering at ÉTS Montréal, enrolled in a dual degree with Arts et Métiers ParisTech. Ranked first in class in French preparatory classes. My research, within the C-PILOT project (LARCASE lab, funded by CRIAQ and NSERC, in partnership with Bombardier), addresses deep learning for multivariate time series: I design complete pipelines, from raw signal to exported model. Three scientific communications accepted or scheduled, including a paper at the AIAA SciTech Forum. An earlier internship in industrial process automation cleared seven weeks of production backlog.

EDUCATION

2025–2027 **École de technologie supérieure (ÉTS)**, Montréal, Canada
M.Sc.A. in Automated Production Engineering, thesis option (in progress)
Cumulative GPA 4.12/4.30. Coursework: SYS819 Deep Learning, SYS843 Neural Networks and Fuzzy Systems (attention architectures and large language models), ENM860 Data Analysis and Predictive Systems (building and deploying predictive systems on a cloud platform), MTR871 Directed Readings.

2023–2027 **Arts et Métiers ParisTech**, Aix-en-Provence, France
General engineering degree (Diplôme d'ingénieur), dual degree with ÉTS (in progress, awarded on completion of the master's)

2020–2023 **Classes préparatoires PTSI / PT***, Lycée Les Eucalyptus, Nice, France
Ranked first in class
Two-year intensive programme in mathematics, physics and engineering science, preparing the national competitive entrance examinations to French engineering grandes écoles.

RESEARCH PROJECT

C-PILOT: behavioral profiling of pilots *September 2025 – August 2027*
Graduate researcher, LARCASE lab, ÉTS, funded by CRIAQ and NSERC, in partnership with Bombardier

- **Problem.** Characterize, without labels and from flight simulator data, the behavioral profiles of pilots during takeoff rotation, then predict the longitudinal control input specific to each profile.
- **Approach.** Design and comparison of four temporal clustering pipelines on 112 variable-length multivariate flight sequences (18 to 81 time steps; masking, padding, phase-wise normalization): variance-weighted DTW with K-medoids, temporal Deep Embedded Clustering, BiLSTM autoencoder with temporal attention, and Transformer encoder with sinusoidal positional encoding. Hyperparameter optimization with Optuna (TPE sampler) on a composite objective combining internal validity and cross-run stability.
- **Innovation.** Regression models specialized per profile: four deep architectures (TCN-LSTM, Transformer, MLP-Mixer, TS-Mixer) compared across three training regimes (global, specialized from scratch, specialized by transfer). Specialization through transfer learning: warm-start from the best fold of the global model with a reduced learning rate, then linear probing with a frozen backbone retraining under 1% of the weights, for an error equivalent to full fine-tuning and a cross-fold standard deviation divided by 5 to 20. Supervised classifier with a reject option based on Mondrian conformal prediction.
- **Results.** Five stable behavioral groups (silhouette 0.608, Davies-Bouldin 0.452, ARI 0.998 over 50 initializations); the choice of sequence encoder dominates the choice of partitioning algorithm on this dataset. Best global model: RMSE 0.0366 and R^2 0.945 with roughly 9,700 parameters; transfer outperforms from-scratch training on all five profiles.
- **Engineering and rigour.** Sliding window, robust per-fold normalization, pilot-stratified cross-validation, ensemble aggregation by fold averaging and ONNX export; classifier evaluated through nested leave-one-pilot-out cross-validation and paired Wilcoxon tests with Holm correction.

PUBLICATIONS AND COMMUNICATIONS

- Neffati, C., Trachet, H., Ghazi, G., and Nyamen Tato, A. A. "Discovering Pilot Behavioral Profiles from Desktop Flight Simulator Data: A Comparison of Temporal Clustering Methods". *AIAA SciTech Forum*, Orlando, 2027. Paper accepted. Largest event in the world for aerospace research, development and technology, with over 6,100 attendees from 48 countries.
- Neffati, C., Trachet, H., Ghazi, G., and Nyamen Tato, A. A. "Temporal Clustering of Flight Simulator Data for

Learner Profiling: A Methodological Comparison". *International Conference on Educational Data Mining (EDM)*, Seoul, 2026. Short paper and poster accepted.

- Neffati, C., Trachet, H., Ghazi, G., and Nyamen Tato, A. A. Oral presentation, *CASI 2026*, Canadian Aeronautics and Space Institute, Winnipeg, October 2026.

PROFESSIONAL EXPERIENCE

Intern, process automation and production scheduling *June – August 2024*
CONTI Monaco, Monaco

- Built a Python solution (Tkinter, PyAutoGUI, OpenPyXL, PyInstaller) driving the AS400 ERP to automate the release, closing and deactivation of production orders. Object-oriented codebase, delivered as an executable with a graphical interface, consistency checks before writing, and an emergency stop.
- Cleared seven weeks of overdue production orders, and moved the plant to releasing orders as soon as the schedule is set.
- Rebuilt the four-month capacity planning tools in Excel and VBA, deriving the minimum headcount per workstation, together with a shortage anticipation system cross-referencing bills of materials against multi-warehouse stock with priority ranking.
- Learned VBA from scratch in ten weeks. Every tool delivered with written operating instructions and user training.

OTHER PROJECTS

Price prediction web application *July 2025*
End-to-end pipeline on a dataset of over 50,000 observations: feature engineering, comparison of linear regression and random forest, then deployment of an interactive Streamlit application hosted on Hugging Face.

Real-time 3D motion capture *January – March 2025*
Tracking of 33 human pose landmarks with OpenCV and MediaPipe, integrated into Unity to animate a synchronized avatar. LISPEN laboratory, Arts et Métiers.

Closed-loop control of a wind turbine nacelle *April – June 2025*
Proportional-integral controller implemented in C++ on an ESP32, system modelling in MATLAB/Simulink with stability analysis, sizing of sensors and actuators.

TECHNICAL SKILLS

Languages	Python, C++, MATLAB, VBA
Machine learning	PyTorch, scikit-learn, Optuna. Sequence and Mixer architectures (BiLSTM, Transformer encoders, TCN-LSTM, MLP-Mixer, TS-Mixer, autoencoders, temporal attention), unsupervised learning, sequence classification and regression, transfer learning (full fine-tuning, linear probing), conformal prediction, ensemble aggregation
Data	NumPy, Pandas, Matplotlib, Seaborn, UMAP, R, Statgraphics. Preprocessing of variable-length signals, masking, normalization, group-wise cross-validation
Deployment	Microsoft Azure, ONNX, Git, Streamlit, Hugging Face, LaTeX
Vision	OpenCV, MediaPipe, Unity
Simulation	MATLAB/Simulink, X-Plane, CATIA V5, Abaqus, ESP32, Arduino

LANGUAGES

French: native • English: professional working proficiency, TOEIC 815, scientific papers written and presented in English • German: B1 • Arabic: conversational